Pavlenko_Chaikin_Assignment_09

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[1]:
```
import plotly.express as px
import plotly.graph_objects as go
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
import numpy as np
```

1 Task 1

1.0.1 Task 1.a

Grid type: uniform Rectilinear Grid

Type of data values: unsigned short with range [0, 4095]

1.0.2 Task 1.b

1.0.3 Task 1.c

2 Task 2

2.0.1 Task 2.a

3 Task 3

3.0.1 Task 3.a

For the given coordinates the following transformation matrix is constructed:

$$\begin{pmatrix} 1 \cdot \frac{1}{\sqrt{2}} & 0 & 1 \cdot \frac{1}{\sqrt{2}} & 1 \\ 0 & 2 \cdot 1 & 0 & -1 \\ 1 \cdot (-\frac{1}{\sqrt{2}}) & 0 & 1 \cdot \frac{1}{\sqrt{2}} & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The transformation matrix from word coordinates back to index coordinates is an inversed transformation matrix

Let us calculate matrices in Python:

[ ]:
```
import numpy as np

transformation_matrix = np.array([
    [1/np.sqrt(2), 0, 1/np.sqrt(2), 1],
    [0, 2, 0, -1],
    [1/np.sqrt(2), 0, -1/np.sqrt(2), 2],
    [0, 0, 0, 1]
])

print(f"Index to world coordinates mapping: \n{transformation_matrix} \n\n␣
 ↪World to index coordinates mapping: \n{np.linalg.
 ↪inv(transformation_matrix)}")
```

```
Index to world coordinates mapping:
[[ 0.70710678  0.          0.70710678  1.        ]
 [ 0.          2.          0.         -1.        ]
 [ 0.70710678  0.         -0.70710678  2.        ]
 [ 0.          0.          0.          1.        ]]

 World to index coordinates mapping:
[[ 0.70710678  0.          0.70710678 -2.12132034]
 [ 0.          0.5         0.          0.5       ]
 [ 0.70710678 -0.         -0.70710678  0.70710678]
 [ 0.          0.          0.          1.        ]]
```

3.0.2 Task 3.b

Use the transformation matrix to determine the world space position of the voxel with index coordinates (2; 1; 5), and the displacement vector in world space that corresponds to incrementing the first index space coordinate by one voxel, the second one by two voxels, and the third one by minus one voxel. Which of the answers changes if we move the origin p and why? (2P)

[ ]:
```
voxel = np.array([2,1,5,1])

e = 1/np.sqrt(2)*np.array([1,0,1])
f = np.array([0,1,0])
g = 1/np.sqrt(2)*np.array([1,0,-1])
ps = [
    np.array([2,1,5,1]),
    np.array([-2,1,5,1]),
    np.array([2,-1,5,1]),
    np.array([2,1,8,1]),
]

for p in ps:
  T = np.array([
    [e[0], f[0], g[0], p[0]],
    [e[1], f[1], g[1], p[1]],
```

```
    [e[2], f[2], g[2], p[2]],
    [0, 0, 0, 1]
  ])
  T_inv = np.linalg.inv(T)

  voxel_world_coords = T @ voxel
  dx = np.array([1,0,0,0])
  dy = np.array([0,2,0,0])
  dz = np.array([0,0,-1,0])

  print(T_inv @ (voxel_world_coords + dx))
  print(T_inv @ (voxel_world_coords + dy))
  print(T_inv @ (voxel_world_coords + dz))
  print("\n\n")
```

```
[2.70710678 1.         5.70710678 1.        ]
[2. 3. 5. 1.]
[1.29289322 1.         5.70710678 1.        ]



[2.70710678 1.         5.70710678 1.        ]
[2. 3. 5. 1.]
[1.29289322 1.         5.70710678 1.        ]



[2.70710678 1.         5.70710678 1.        ]
[2. 3. 5. 1.]
[1.29289322 1.         5.70710678 1.        ]



[2.70710678 1.         5.70710678 1.        ]
[2. 3. 5. 1.]
[1.29289322 1.         5.70710678 1.        ]
```

None of the displaced voxel is changed when the origin p is moved, because this vector stands for translation in the world coordinates and is complemented when we multiply with the inversed transformation when going back to the index space.

4 Task 4

4.1 Task 4.a

To analyze the accuracy of the **forward finite difference** method, we apply a Taylor expansion of (f(x+h)) around (x):

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f^{(3)}(x)}{6}h^3 + \mathcal{O}(h^4)$$

Substitute into the forward difference:

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{f''(x)}{2}h + \frac{f^{(3)}(x)}{6}h^2 + \mathcal{O}(h^3)$$

The error starts at order (h), so this scheme is **first-order accurate**.

4.2 Task 4.b

Now consider the central difference:

$$\frac{f(x+h) - f(x-h)}{2h}$$

Using Taylor expansions:

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \frac{f^{(3)}(x)}{6}h^3 + \frac{f^{(4)}(x)}{24}h^4 + \mathcal{O}(h^5)$$

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)}{2}h^2 - \frac{f^{(3)}(x)}{6}h^3 + \frac{f^{(4)}(x)}{24}h^4 + \mathcal{O}(h^5)$$

Subtract and divide:

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{f^{(3)}(x)}{6}h^2 + \mathcal{O}(h^4)$$

This means the central scheme is **second-order accurate**.

5 Task 5

5.1 Task 5.a

The standard bilinear interpolation is:

$$f(x,y) = f_{i,j}(1-x)(1-y) + f_{i+1,j}x(1-y) + f_{i,j+1}(1-x)y + f_{i+1,j+1}xy$$

This can be rewritten as:

$$f(x,y) = \eta \cdot x \cdot y + (f_{i+1,j} - f_{i,j})x + (f_{i,j+1} - f_{i,j})y + f_{i,j}$$

where:

$$\eta = f_{i,j} - f_{i+1,j} - f_{i,j+1} + f_{i+1,j+1}$$

From this, we get:

$$y_0 = -\frac{f_{i+1,j} - f_{i,j}}{\eta}, \quad x_0 = -\frac{f_{i,j+1} - f_{i,j}}{\eta}$$

And:

$$\gamma = f_{i,j} - \eta x_0 y_0 = \frac{f_{i,j}f_{i+1,j+1} - f_{i+1,j}f_{i,j+1}}{\eta}$$

5.2 Task 5.b

We only divide by η, so division by zero is possible **only if** $\eta = 0$.
But in the cases where the asymptotic decider is used, $\eta \neq 0$ because:

$$f_{i,j} + f_{i+1,j+1} \neq f_{i+1,j} + f_{i,j+1}$$

Hence, division by zero **does not occur** in relevant cases.

5.3 Task 5.c

The paper defines a similar value $B(S_\alpha, T_\alpha)$ as:

$$\frac{B_{00}B_{11} + B_{10}B_{01}}{B_{00} + B_{11} - B_{01} - B_{10}}$$

But this is incorrect. The correct numerator should be:

$$B(S_\alpha, T_\alpha) = \frac{B_{00}B_{11} - B_{10}B_{01}}{B_{00} + B_{11} - B_{01} - B_{10}}$$

So the mistake is the **plus sign** in the numerator; it should be a **minus**.

6 Task 6

- **First cube** (left): corresponds to base case 4, ambiguous on the top face.

$$\gamma = \frac{1 \cdot 1 - (-2) \cdot (-2)}{1 + 1 - (-2) - (-2)} = -\frac{1}{2} \leq 0$$

So, the two negative vertices are **connected**.

- **Second cube**: same base case, different values.

$$\gamma = \frac{2 \cdot 2 - (-1) \cdot (-1)}{2 + 2 - (-1) - (-1)} = \frac{1}{2} > 0$$

Here, the negative vertices are **disconnected**.

- **Third cube**: symmetric version of base case 8 with ambiguity on all faces.

$$\gamma = \frac{1 \cdot 1 - (-2) \cdot (-2)}{1 + 1 - (-2) - (-2)} = -\frac{1}{2} \leq 0$$

All negative vertices are **connected** due to symmetry and (0).